COMPUTATIONAL SCIENCES
WINTER TERM 2026
FREIE UNIVERSITÄT BERLIN

Report

Incompressible Navier Stokes Project

Annie, Bianca, Ceren, Gotkug, Jaza, Louis

Gutachter(in):	Dr. Luca Donati
Semester:	Winter term 2026
Verfasser:	Vorname Nachname
Matrikel-Nr.:	1234567
Adresse:	Straße Nr, PLZ Ort
Email:	mail@mail.de
Telefon:	030-123 456 78
Studienfach:	MSc. Computational Sciences
Abgabetermin:	**01. Januar 2099**

Abstract

TEXT

Contents

List of Figures

List of Tables

Clymo (1970)

1 Introduction

1.1 Field of relevance

TEXT

1.2 Research goal

TEXT

1.3 Structure of this report

TEXT

2 "‘Methods"’: Konzept, Analyse, Experiment etc.

2.1 First task

2.1.1 Model

explain problem and model

2.1.2 Operator Splitting

Consider a typical PDE of structure

$$\partial_t \boldsymbol{u} = (A + B + C) \cdot \boldsymbol{u}$$

Here, $\boldsymbol{u}$ denotes a vector variable, e.g. the velocity in a fluid at a certain location . A, B, C on the other hand are so called *operators*. Operators are terms in the PDE and can describe certain physical processes. For instance, consider the basic Navier-Stokes equation

$$\partial_t T = \Delta T - \boldsymbol{u} \nabla T$$

. Here, we could set $A := \Delta, B := -\boldsymbol{u}\nabla$ and obtain:

$$\partial_t T = (A - B)T$$

A expresses the , B represents advection .

what does it express?

Usually, numerical methods have to be applied to solve such a problem of partial differential equations. As these are often computationally expensive *operator splitting* provides a procedure to reduce costs for the solution of the PDE. We "split" the problem into subproblems, meaning that we consider each operator individually:

clarify, whether we really have this problem?

$$\partial_t \boldsymbol{u} = A\boldsymbol{u} \tag{2.1}$$
$$\partial_t \boldsymbol{u} = B\boldsymbol{u} \tag{2.2}$$

Here, we numerically solve one of the the two equations and use the result as the initial conditions for the second to solve it. This is repeated for all timesteps. Of course, this produces an error, as the new initial conditions for the second equation do not completely coincide with the initial conditions we used for the first equation. Thus, solving by integrating the operators subsequently creates an error. Here, different methods of operator splitting exist and different errors are the consequence. In general, the error remains small if the timestep Δt is chosen small in the numerical integration scheme.

First order () operator splitting

include author

First order operator splitting means to solve both operators subsequently for a given timestep Δt. To further investigate this, consider

$$\partial_t \boldsymbol{u} = (A + B)\boldsymbol{u} \tag{2.3}$$
$$\Rightarrow \partial_t \boldsymbol{u} = A\boldsymbol{u} \tag{2.4}$$
$$\partial_t \boldsymbol{u} = B\boldsymbol{u} \tag{2.5}$$
$$\tag{2.6}$$

with A, B being again some operators. For our problem, these operators will be part of the advection-diffusion equation, as we see later. Note, that we derived two **linear** problems after we applied operator splitting.

That is why we can find simple solutions to the formulated splitted operator problems, namely

$$u_A(t) = e^{(t-t_0)A}u_0 \tag{2.7}$$
$$u_B(t) = e^{(t-t_0)B}u_0 \tag{2.8}$$

In our case we only want to solve for the next discrete time interval $[t_k, t_{k+1}] = [t_k, t_k + \Delta t]$ – not the whole time interval. That is why we modify the solutions for the first order operator splitting and rewrite them to

$$u_{k+1,A} = e^{(t_{k+1}-t_k)A}u_{k,A} \tag{2.9}$$
$$u_{k+1,B} = e^{(t_{k+1}-t_k)B}u_{k,B} \tag{2.10}$$

. With $\Delta t = t_{k+1} - t_k$ we finally get the solution for each operator with respect to the discetized time step Δt:

$$u_{k+1,A} = e^{\Delta t A} u_{k,A} \tag{2.11}$$
$$u_{k+1,B} = e^{\Delta t B} u_{k,B} \tag{2.12}$$

.

For first-order operator splitting the following algorithm is applied to the abstract problem $\partial_t \boldsymbol{u} = (A + B)\boldsymbol{u}$ is split as shown into the two seperate operator problems:

1. Set $\boldsymbol{u}_{k,A} := \boldsymbol{u}_k$
2. Solve $\boldsymbol{u}_{k+1,A} = e^{\Delta t A}\boldsymbol{u}_{k,A}$ on the next interval $[t_k, t_k + \Delta t]$.
3. We obtain $\boldsymbol{u}_{k+1,A}$.
4. Consider $\boldsymbol{u}_{k+1,A}$ as the new initial condition in the second operator equation: $\boldsymbol{u}_{k,B} := \boldsymbol{u}_{k+1,A}$
5. Solve $\boldsymbol{u}_{k+1,B} = e^{\Delta t B}\boldsymbol{u}_{k,B} = e^{\Delta t B}\boldsymbol{u}_{k+1,A}$.
6. We obtain $\boldsymbol{u}_{k+1,B}$
7. We consider $\boldsymbol{u}_{k+1,B}$ as our solution for $\boldsymbol{u}_{k+1}$

Written in a compact manner, we obtain

$$u_{k+1} = e^{\Delta t B} e^{\Delta t A} u_k$$

as our first-order operator splitting procedure. Like already mentioned, this algorithm relies on the assumption, that for small timesteps Δt only small difference remains between $u_{k+1,A}$ and $u_{k+1,B}$ when evaluated subsequently. This algorithm is simple and easy to implement. Nevertheless, the GTE error is of order $\mathcal{O}(\Delta t)$ () and can be further reduced. This leads to the second-order operator methods.

include some computational efficiency stuff, maybe even related to our testing?

include source

Second order operator splitting (Strang-splitting)

To reduce the GTE of the operator splitting scheme to an error of order $\mathcal{O}(\Delta t^2)$ the following can be applied to approximate the solution $\boldsymbol{u}_{k+1}$ at the next timestep $t_{k+1} = t_k + \Delta t$:

$$\boldsymbol{u}_{k+1} = e^{\frac{\Delta t A}{2}} \cdot e^{\Delta t B} \cdot e^{\frac{\Delta t A}{2}} \boldsymbol{u}_k \tag{2.13}$$

2.1.3 Application of operator splitting on the Velocity equation

is it really the name?

We now apply operator splitting in the context of the velocity equation

$$\partial_t \boldsymbol{u} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} + \frac{1}{\rho} \nabla p = \nu \Delta \boldsymbol{u} + \beta T \boldsymbol{e}_z$$

and obtain three different operators:

1. Advection: $\partial_z \boldsymbol{u} = \boldsymbol{u} \cdot \nabla \boldsymbol{u}$
2. Diffusion + forcing: $\partial_t \boldsymbol{u} = \nu \Delta \boldsymbol{u} + \beta T_{\boldsymbol{e}_z}$
3. Projection step (Incompressibility): $\boldsymbol{u}^{k+1} = \boldsymbol{u}^* - \Delta \boldsymbol{p}, \quad$ with $\nabla \cdot \boldsymbol{u}^{k+1} = 0$
   The pressure acts like a Lagrangian multiplier and enforces $\nabla \boldsymbol{u} = 0$

1. **Step 1:**
   Ignore Incompressibility: $\partial_t \boldsymbol{u} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = \nu \cdot \Delta \boldsymbol{u} + f$. Here, we obtain the intermediate velocity $\boldsymbol{u}^* \Rightarrow \nabla \cdot \boldsymbol{u} \neq 0$
2. **Step 2:**
   We now use this intermediate velocity to respect incompressibility and include pressure:

   $$\boldsymbol{u}^{k+1} = \boldsymbol{u}^* - \nabla \boldsymbol{p}$$

   We use the *Helmholtz decomposition* to do this:

   $$\boldsymbol{u}^* = \boldsymbol{u}^{\perp} + \nabla \phi$$

   It follows immediately, that $\boldsymbol{u}^{k+1} = \boldsymbol{u}^{\perp}$. We want

   $$\nabla \boldsymbol{u}_{k+1} = 0 \Rightarrow \nabla(\boldsymbol{u}^* - \nabla \boldsymbol{p}) = 0 \Rightarrow \nabla(\boldsymbol{u}^* - \nabla \boldsymbol{p}) = 0 \Rightarrow \nabla \boldsymbol{u}^* = \Delta p$$

   as this gives us the Poisson equation for pressure.

Now we want to apply operator splitting for all the different terms. We begin with the advection operator term.

$$\partial_t \boldsymbol{u} + a_y \partial_y \boldsymbol{u} + a_z \partial_z \boldsymbol{u} = 0$$

Here, we defined $\boldsymbol{a} = \begin{pmatrix} a_y(y,z) \\ a_z(y,z) \end{pmatrix}$ and $\boldsymbol{u} = \begin{pmatrix} v \\ w \end{pmatrix}$

The actual splitting is now exececuted in space first. The two spatial operators for each spatial dimension lead to the following approach:

1. Solve
$$\partial_t \boldsymbol{u} + a_y \partial_y \boldsymbol{u} = 0$$
Or, more explicitly written:
$$\begin{cases} \partial_t v + a_y \partial_y v & = 0 \\ \partial_t w + a_y \partial_y w & = 0 \end{cases}$$
We do the same for the second spatial dimension:
2. Solve
$$\partial_t \boldsymbol{u} + a_z \partial_z \boldsymbol{u} = 0$$
Again, more explicitly written:
$$\begin{cases} \partial_t v + a_z \partial_z \boldsymbol{u} & = 0 \\ \partial_t w + a_z \partial_z w & = 0 \end{cases}$$
3. Apply Lie Splitting in time: $\boldsymbol{u}^{k+1} = A_z(\Delta t) A_y(\Delta t) \boldsymbol{u}^k$

2.1.4 Upward Finite Difference Method

To introduce the Upward Finite Difference Method we first want to look again at the one-dimensional linear advection equation

$$\frac{\partial_u}{\partial_t} + a\frac{\partial u}{\partial x} = 0$$

Intuitively, this equation describes a wave that propagates along the x-axis with velocity a. Now imagine the x-axis laid out horizontally in fron of you. If $a > 0$, the wave propagates to the right. Thus, an upwind direction would correspond to the direction in the left – towards smaller values. For $a < 0$ this changes accordingly: The upwind direction corresponds to the direction of higher x-values, so to the right.

That is why we will use backward finite difference methods for $a > 0$ and forward finite difference methods for $a < 0$. We will use the upwind discretization in a onedimensional case and therefore discretize the grid in the following manner:

$$\begin{aligned} x_i &:= i\,\Delta x, \quad i = 0, \dots, N \\ t^n &:= n\,\Delta t \end{aligned}$$

As already described, we will use different methods depending on the exact value of a to ensure the upwind finite difference scheme.

also define discrete no of steps for Time? So maybe N_x and N_t?

1. $a > 0$ (backward):
$$\partial_x \boldsymbol{u} \approx \frac{\boldsymbol{u}_i - \boldsymbol{u}_{i-1}}{\Delta x}$$
2. $a < 0$ (forward):
$$\partial_x \boldsymbol{u} \approx \frac{\boldsymbol{u}_{i+1} - \boldsymbol{u}_i}{\Delta x}$$

After executing th step for the spatial dimensions, we proceed with the discrete step in time:

$$\frac{\boldsymbol{u}_{i+1}^n - \boldsymbol{u}_i^{n+1}}{\Delta t} = -a(\partial_x \boldsymbol{u})_i^n$$

1. $a > 0$ (backward):

$$\begin{aligned} \frac{\boldsymbol{u}_{i+1}^n - \boldsymbol{u}_i^{n+1}}{\Delta t} &= -a(\partial_x \boldsymbol{u})_i^n && (2.14) \\ \Rightarrow \boldsymbol{u}_i^{n+1} - \boldsymbol{u}_i^n &= -a\frac{\Delta t}{\Delta x}(\boldsymbol{u}_i^n - \boldsymbol{u}_{i-1}^n) && (2.15) \\ \Rightarrow \boldsymbol{u}_i^{n+1} &= \left(1 - a\frac{\Delta t}{\Delta x}\right)\boldsymbol{u}_i^n + a\frac{\Delta t}{\Delta x}\boldsymbol{u}_{i-1}^n && (2.16) \\ \Rightarrow \boldsymbol{u}_i^{n+1} &= A_1 \boldsymbol{u}_i^n && (2.17) \end{aligned}$$

2. $a < 0$ (forward):

$$\begin{aligned} \frac{\boldsymbol{u}_{i+1}^n - \boldsymbol{u}_i^{n+1}}{\Delta t} &= -a(\partial_x \boldsymbol{u})_i^n && (2.18) \\ \Rightarrow \boldsymbol{u}_i^{n+1} - \boldsymbol{u}_i^n &= -a\frac{\Delta t}{\Delta x}(\boldsymbol{u}_{i+1}^n - \boldsymbol{u}_i^n) && (2.19) \\ \Rightarrow \boldsymbol{u}_i^{n+1} &= \left(1 - a\frac{\Delta t}{\Delta x}\right)\boldsymbol{u}_i^n - a\frac{\Delta t}{\Delta x}\boldsymbol{u}_{i+1}^n && (2.20) \\ \Rightarrow \boldsymbol{u}_i^{n+1} &= A_2 \boldsymbol{u}_i^n && (2.21) \end{aligned}$$

Thus, we obtained our operator for each upwind case which depends only on the sign of a. To shorten the emerging equations for all of our discretized locations we will write the operator in a compact matrix notation.

$$A_1 = \begin{pmatrix} \left(1 - a\frac{\Delta t}{\Delta x}\right) & 0 & & \dots & 0 \\ a\frac{\Delta t}{\Delta x} & \ddots & & & \vdots \\ 0 & \ddots & \ddots & & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \dots & 0 & a\frac{\Delta t}{\Delta x} & \left(1 - a\frac{\Delta t}{\Delta x}\right) \end{pmatrix}$$

$$A_2 = \begin{pmatrix} \left(1 - a\frac{\Delta t}{\Delta x}\right) & -a\frac{\Delta t}{\Delta x} & 0 & \dots & 0 \\ 0 & \ddots & \ddots & & \vdots \\ & & \ddots & \ddots & 0 \\ \vdots & & & \ddots & -a\frac{\Delta t}{\Delta x} \\ 0 & \dots & & 0 & \left(1 - a\frac{\Delta t}{\Delta x}\right) \end{pmatrix}$$

Stability of the Operators

In general, the stability condition for a matrix operator A is given by $\|A\|_\infty \leq 1$. We will investigate our both operators A_1 (corresponds to $a > 0$ and therefore backward finite difference method) and A_2 (corresponds to $a < 0$ and therefore forward finite difference method) with respect to this stability condition.

$$\|A_1\|_\infty = \left|a\frac{\Delta t}{\Delta x}\right| + \left|1 - a\frac{\Delta t}{\Delta x}\right| \leq 1 \tag{2.22}$$

$$\Rightarrow^{a>0} \left|1 - a\frac{\Delta t}{\Delta x}\right| \leq 1 - a\frac{\Delta t}{\Delta x} \tag{2.23}$$

$$\Rightarrow 1 - a\frac{\Delta t}{\Delta x} \geq 0 \tag{2.24}$$

$$\Rightarrow a\frac{\Delta t}{\Delta x} \leq 1 \tag{2.25}$$

Repeating this for our second operator A_2 we get

$$\|A_2\|_\infty = \left|a\frac{\Delta t}{\Delta x}\right| + \left|1 - a\frac{\Delta t}{\Delta x}\right| \leq 1 \tag{2.26}$$

$$\Rightarrow^{a>0} \left|1 - a\frac{\Delta t}{\Delta x}\right| \leq 1 + a\frac{\Delta t}{\Delta x} \tag{2.27}$$

$$\Rightarrow 1 + a\frac{\Delta t}{\Delta x} \geq 0 \tag{2.28}$$

$$\Rightarrow -a\frac{\Delta t}{\Delta x} \leq 1 \tag{2.29}$$

and, thus, obtain

$$\left|a\frac{\Delta t}{\Delta x}\right| \leq 1$$

as our general stability condition for our operator.

2.1.5 Twodimensional upwind scheme with operator splitting

We now want to jointly apply operator splitting and the upwind finite difference method on our twodimensional advection equation. Therefore, we define $u_{ij}^n := u(t^n, y_i, z_i)$ and assume an equidistant grid as the discretized version of our problem. Therefore we declare Δy and Δz as the distances between discrete grid points in space.

We first want to solve the advection term in the y direction. We fix a value z_j and solve the resulting equation

$$\partial_t \boldsymbol{u} + a_y \partial_y \boldsymbol{u} = 0$$

Whether forward or backward finite difference method is applied depends on a again:

$$(\partial_y \boldsymbol{u})_{ij} = \begin{cases} \frac{\boldsymbol{u}_{ij} - \boldsymbol{u}_{i-1\,j}}{\Delta y} & a_y > 0 \\ \frac{\boldsymbol{u}_{i+1\,j} - \boldsymbol{u}_{i\,j}}{\Delta y} & a_y < 0 \end{cases}$$

As a result of our first operator we obtain

$$u_{ij}^* = u_{ij}^n - \Delta t a_{y,ij} (\partial_y \boldsymbol{u})_i j$$

where we ensure

$$a_y \frac{\Delta t}{\Delta y} \leq 1$$

. We use this result to proceed with the solution of the z-coordinate corresponding to the second operator:

$$\partial_t \boldsymbol{u} + a_z \partial_z \boldsymbol{u} = 0$$

We define analogously

$$(\partial_y \boldsymbol{u}^*)_{ij} = \begin{cases} \frac{\boldsymbol{u}^*_{ij} - \boldsymbol{u}^*_{i-1\,j}}{\Delta y} & a_y > 0 \\ \frac{\boldsymbol{u}^*_{i+1\,j} - \boldsymbol{u}^*_{i\,j}}{\Delta y} & a_y < 0 \end{cases}$$

Using the result from the first splitted operator we are now able to compute

$$u_{ij}^{n+1} = u_{ij}^* - \Delta t a_{z,ij} (\partial_z \boldsymbol{u})_i j$$

We ensure stability in this case as well by respecting $\frac{|a_z|\Delta t}{\Delta z} \leq 1$.

We can now derive a stability condition for all of our discretized points in space. We first obtain

$$\max |a_y| = \max_{i,j} |a_y(y_i, z_j)|$$

and

$$\max |a_z| = \max_{i,j} |a_z(y_i, z_j)|$$

. By the help of these variables we can now deduce the overall constraint for Δt, that is neccessary to ensure stability:

$$\Delta t \leq \min \left(\frac{\Delta y}{\max |a_y|}, \frac{\Delta z}{\max |a_z|} \right)$$

Summarized, we successfully implemented operator splitting and upwind finite differende method for the linear advection term and ensured stability all over our domain.

boundary conditions implementation

2.2 Incompressible flow – Chorins projection method

We consider the Navier-Stokes equations for velocity

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} + \frac{1}{\rho} \nabla p = \nu \Delta \mathbf{u} \qquad \text{with } \beta = 0 \tag{2.30}$$

$$\nabla \cdot \mathbf{u} = 0 \tag{2.31}$$

Here, we implement the boundary condition

$$\mathbf{u}(t, 0, z) = \mathbf{u}(t, 1, z) \tag{2.32}$$

$$\mathbf{w}(t, y, 0) = \mathbf{w}(t, y, 1) = 0 \qquad \forall t \in [0, t_{end}], (y, z) \in \Omega \tag{2.33}$$

In order to solve this differential equation we use a projection method. . As demanded by the project task we implement Chorins projection method. Conceptually, this approach can be split into two steps:

explain why we use a projection method

1. We calculate an intermediate velocity that does not satisfy the incompressibility constraint ($\nabla \cdot \mathbf{u} = 0$). This intermediate velocity is computed at each time step (ignoring the pressure term)

2. Subsequently, the pressure p is used to *project* the intermediate velocity onto a space of divergent free velocity field.

Formally, we perform

1.
$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} = -(\mathbf{u}^n \cdot \nabla)\mathbf{u}^n + \nu \nabla^2 \mathbf{u}^n$$

2.
$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^*}{\Delta t} = -\frac{1}{\rho} \nabla p^{n+1}$$

Two things are especially noteworthy about this procedure. Firstly, the pressure p is time-invariant. That menas, that the pressure p is unaffected by our timestep Δt. Secondly, we can summarize that this procedure is a kind of operator splitting.

But we are not finished here. In our first step of the described scheme we obtain the intermediate velocity $\mathbf{u}^*$. However, this velocity vector is not complying with our constraint of zero divergence. Additionally, the solution of the second equation of this scheme requires knowledge about p. That is why we rewrite the second equation:

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla p^{n+1}$$

Ensuring the absence of divergence ($\nabla \mathbf{u}^{n+1} = 0$) allows us to further extend this equation:

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla p^{n+1} \tag{2.34}$$

$$\text{respecting divergence} \quad 0 = \nabla \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla^2 p^{n+1} \tag{2.35}$$

$$\Rightarrow \Delta p^{n+1} = \frac{\rho}{\Delta t} (\nabla \mathbf{u}^*) \tag{2.36}$$

The last result refelcts the Poisson equation for pressure.

We can see that the method is called a 'projection' because we decompose $\mathbf{u}^*$ into a divergence-free (solenoidal) part $\mathbf{u}_{sol}$ and irraotational part $\mathbf{u}_{\text{irrot}}$. Then with the projection step $\mathbf{u}_{irrot} = \nabla \phi$ is removed so that the remaining part is negative divergence-free.

$$\mathbf{u} = \mathbf{u}_{\text{sol}} + \mathbf{u}_{\text{irrot}} = \mathbf{u}_{\text{sol}} + \nabla \phi$$

Since $\nabla \mathbf{u}_{\text{sol}} = 0$ the whole expression simplifies when we calculate the divergence of the total expression:

$$\Delta \mathbf{u} = \Delta \phi$$

Here, $\Delta = \nabla^2$ represents the Laplacian operator. Consequently, the solenoidal part can be calculated as $\mathbf{u}_{\text{sol}} = \mathbf{u} - \nabla \phi$.

This kind of projection method is called Helmholtz-Hodge decomposition.

We once again consider the Poisson equation we want to solve:

$$\Delta p = \frac{\rho}{\Delta t} (\nabla \mathbf{u}^*) \quad \text{in the domain } \Omega = [0, 1]^2$$

If we assume Neumann boundary conditions it leads to

$$\frac{\partial p}{\partial \mathbf{n}} = 0 \Leftrightarrow \nabla p \cdot \mathbf{n} = 0 \quad \text{on } \partial \Omega$$

With this boundary condition we get a solution that is not unique as p is defined up to an additive constant. To solve the Poisson equation we equidistantly discretize the spatial domain.

$$y_i = i \Delta y \quad i = 0, ..., N_y - 1$$

Boundary	BC	numerical BC	Resulting condition
Bottom$(z = 0, j = 0)$	$\frac{\partial p}{\partial z} = 0$	$\frac{p_{i,1} - p_{i,-1}}{2\Delta z} = 0$	$p_{i,1} = p_{i,-1}$
Top$(z = 1, j = N_z - 1)$	$\frac{\partial p}{\partial z} = 0$	$\frac{p_{i,N_z} - p_{i,-2}}{2\Delta z} = 0$	$p_{i,N_z} = p_{i,N_z-2}$
Left$(y = 0, i = 0)$	$\frac{\partial p}{\partial y} = 0$	$\frac{p_{1,j} - p_{-1,j}}{2\Delta z} = 0$	$p_{1,j} = p_{-1,j}$
Right$(y = 1, i = N_y - 1)$	$\frac{\partial p}{\partial y} = 0$	$\frac{p_{N_y,j} - p_{N_y-2,j}}{2\Delta z} = 0$	$p_{N_y,j} = p_{N_y-2,j}$

Table 2.1: **Dsicretized Boundary Conditions (BC)** Boundary conditions for all of the four boundaries of the twodimensional domain $\Omega = [0, 1]^2$. Each row represents one boundary and shows the continous and numerical representation of the boundary conditions given in the task. Numerical approximation was performed by using a central finite difference approach.

$$z_j = j\Delta z \quad j = 0, ..., N_z - 1$$

To approximate the second derivatives (Laplace operator) we implement the second finite difference in two dimensions:

$$\Delta p = \nabla^2 p = \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2} \approx \frac{p_{i+1,j} - 2p_{i,j} + p_{i-1,j}}{\Delta y^2} + \frac{p_{i,j+1} - 2p_{i,j} + p_{i,j-1}}{\Delta z^2}$$

2.2.1 Discretized Poisson equation

We now plug the discretized Laplace operator into our Poisson equation. Remember that the Poisson we want to solve was

$$\Delta p^{n+1} = \frac{rho}{\Delta t}(\nabla \mathbf{u}^*)$$

Thus, plugging in the numerical approximations and rearranging yields:

$$-2(\frac{1}{\Delta y^2} + \frac{1}{\Delta z^2})p_{i,j} + \frac{p_{i+1,j} + p_{i-1,j}}{\Delta y^2} + \frac{p_{i,j+1} + p_{i,j-1}}{\Delta z^2} = \frac{\rho}{\Delta t}(\nabla \mathbf{u}^*)_{i,j}$$

We also have to ensure, that our numerical solution respects the boundary conditions. We will use a central finite difference approach for this. The central finite difference can generally be written as

$$f_i' \approx \frac{f_{i+1} - f_{i-1}}{2\Delta x}$$

Each boundary of our domain then corresponds to another numerical condition to be met which can be seen in table 2.2.1.

2.2.2 Numerical procedure

Summarising, we obtain the following numerical procedure:

1. Solve $\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} = -(\mathbf{u}^n \cdot \nabla) \cdot \mathbf{u}^n + \nu \Delta \mathbf{n}^n$ with a suitable integrator scheme
2. Obtain p by solving the discretized Poisson equation
3. Solve $\frac{\mathbf{u}^{n+1} - \mathbf{u}^*}{\Delta t} = -\frac{1}{\rho} \nabla p$ and obtain $\mathbf{u}^{n+1}$

2.3 Further tasks

3 Ergebnisse

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem. Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu. In enim justo, rhoncus ut, imperdiet a, venenatis vitae, justo. Nullam dictum felis eu pede mollis pretium. Integer tincidunt. Cras dapibus. Vivamus elementum semper nisi. Aenean vulputate eleifend tellus. Aenean leo ligula, porttitor eu, consequat vitae, eleifend ac, enim. Aliquam lorem ante, dapibus in, viverra quis, feugiat a, tellus. Phasellus viverra nulla ut metus varius laoreet. Quisque rutrum. Aenean imperdiet. Etiam ultricies nisi vel augue. Curabitur ullamcorper ultricies nisi. Nam eget dui. Etiam rhoncus. Maecenas tempus, tellus eget condimentum rhoncus, sem quam semper libero, sit amet adipiscing sem neque sed ipsum. Nam quam nunc, blandit vel, luctus pulvinar, hendrerit id, lorem. Maecenas nec odio et ante tincidunt tempus. Donec vitae sapien ut libero venenatis faucibus. Nullam quis ante.

3.1 Erster Abschnitt

Etiam sit amet orci eget eros faucibus tincidunt. Duis leo. Sed fringilla mauris sit amet nibh. Donec sodales sagittis magna. Sed consequat, leo eget bibendum sodales, augue velit cursus nunc, quis gravida magna mi a libero. Fusce vulputate eleifend sapien. Vestibulum purus quam, scelerisque ut, mollis sed, nonummy id, metus. Nullam accumsan lorem in dui. Cras ultricies mi eu turpis hendrerit fringilla. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; In ac dui quis mi consectetuer lacinia. Nam pretium turpis et arcu. Duis arcu tortor, suscipit eget, imperdiet nec, imperdiet iaculis, ipsum. Sed aliquam ultrices mauris. Integer ante arcu, accumsan a, consectetuer eget, posuere ut, mauris. Praesent adipiscing. Phasellus ullamcorper ipsum rutrum nunc. Nunc nonummy metus. Vestibulum volutpat pretium libero. Cras id dui. Aenean ut eros et nisl sagittis vestibulum. Nullam nulla eros, ultricies sit amet, nonummy id, imperdiet feugiat, pede. Sed lectus. Donec mollis hendrerit risus. Phasellus nec sem in justo pellentesque facilisis. Etiam imperdiet imperdiet orci. Nunc nec neque. Phasellus leo dolor, tempus non, auctor et, hendrerit quis, nisi. Curabitur ligula sapien, tincidunt non, euismod vitae, posuere imperdiet, leo. Maecenas malesuada. Praesent congue erat at massa. Sed cursus turpis vitae tortor. Donec posuere vulputate arcu. Phasellus accumsan cursus velit. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Sed aliquam, nisi quis porttitor congue, elit erat euismod orci, ac placerat dolor lectus quis orci. Phasellus consectetuer vestibulum elit. Aenean tellus metus, bibendum sed, posuere ac, mattis non, nunc. Vestibulum fringilla pede sit amet augue. In turpis. Pellentesque posuere. Praesent turpis. Aenean posuere, tortor sed cursus feugiat, nunc augue blandit nunc, eu sollicitudin urna dolor sagittis lacus. Donec elit libero, sodales nec, volutpat a, suscipit non, turpis. Nullam sagittis. Suspendisse pulvinar, augue ac venenatis condimentum, sem libero volutpat nibh, nec pellentesque velit pede quis nunc. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Fusce id purus. Ut varius tincidunt libero. Phasellus dolor. Maecenas vestibulum mollis diam. Pellentesque ut neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. In dui magna, posuere eget, vestibulum et, tempor auctor, justo. In ac felis quis tortor malesuada pretium. Pellentesque auctor neque nec urna. Proin sapien ipsum, porta a, auctor quis, euismod ut, mi. Aenean viverra rhoncus pede. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Ut non enim eleifend felis pretium feugiat. Vivamus quis mi. Phasellus a est. Phasellus magna. In hac habitasse platea dictumst. Curabitur at lacus ac velit ornare lobortis. Curabitur a felis in nunc fringilla tristique. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem. Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu. In enim justo, rhoncus ut, imperdiet a, venenatis vitae, justo. Nullam dictum felis eu pede mollis pretium. Integer tincidunt. Cras dapibus. Vivamus elementum semper nisi. Aenean vulputate eleifend tellus. Aenean leo ligula, porttitor eu, consequat vitae, eleifend ac, enim. Aliquam lorem ante, dapibus in, viverra quis, feugiat a, tellus. Phasellus viverra nulla ut metus varius laoreet. Quisque rutrum. Aenean imperdiet. Etiam ultricies nisi vel augue. Curabitur ullamcorper ultricies nisi. Nam eget dui. Etiam rhoncus...

3.2 Zweiter Abschnitt

Maecenas tempus, tellus eget condimentum rhoncus, sem quam semper libero, sit amet adipiscing sem neque sed ipsum. Nam quam nunc, blandit vel, luctus pulvinar, hendrerit id, lorem. Maecenas nec odio et ante tincidunt tempus. Donec vitae sapien ut libero venenatis faucibus. Nullam quis ante. Etiam sit amet orci eget eros faucibus tincidunt. Duis leo. Sed fringilla mauris sit amet nibh. Donec sodales sagittis magna. Sed consequat, leo eget bibendum sodales, augue velit cursus nunc, quis gravida magna mi a libero. Fusce vulputate eleifend sapien. Vestibulum purus quam, scelerisque ut, mollis sed, nonummy id, metus. Nullam accumsan lorem in dui. Cras ultricies mi eu turpis hendrerit fringilla. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; In ac dui quis mi consectetuer lacinia. Nam pretium turpis et arcu. Duis arcu tortor, suscipit eget, imperdiet nec, imperdiet iaculis, ipsum. Sed aliquam ultrices mauris. Integer ante arcu, accumsan a, consectetuer eget, posuere ut, mauris. Praesent adipiscing. Phasellus ullamcorper ipsum rutrum nunc. Nunc nonummy metus. Vestibulum volutpat pretium libero. Cras id dui. Aenean ut eros et nisl sagittis vestibulum. Nullam nulla eros, ultricies sit amet, nonummy id, imperdiet feugiat, pede. Sed lectus. Donec mollis hendrerit risus. Phasellus nec sem in justo pellentesque facilisis. Etiam imperdiet imperdiet orci. Nunc nec neque. Phasellus leo dolor, tempus non, auctor et, hendrerit quis, nisi. Curabitur ligula sapien, tincidunt non, euismod vitae, posuere imperdiet, leo. Maecenas malesuada. Praesent congue erat at massa. Sed cursus turpis vitae tortor. Donec posuere vulputate arcu. Phasellus accumsan cursus velit. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Sed aliquam, nisi quis porttitor congue, elit erat euismod orci, ac placerat dolor lectus quis orci. Phasellus consectetuer vestibulum elit. Aenean tellus metus, bibendum sed, posuere ac, mattis non, nunc. Vestibulum fringilla pede sit amet augue. In turpis. Pellentesque posuere. Praesent turpis. Aenean posuere, tortor sed cursus feugiat, nunc augue blandit nunc, eu sollicitudin urna dolor sagittis lacus. Donec elit libero, sodales nec, volutpat a, suscipit non, turpis. Nullam sagittis. Suspendisse pulvinar, augue ac venenatis condimentum, sem libero volutpat nibh, nec pellentesque velit pede quis nunc. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Fusce id purus. Ut varius tincidunt libero. Phasellus dolor. Maecenas vestibulum mollis diam. Pellentesque ut neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. In dui magna, posuere eget, vestibulum et, tempor auctor, justo. In ac felis quis tortor malesuada pretium. Pellentesque auctor neque nec urna. Proin sapien ipsum, porta a, auctor quis, euismod ut, mi. Aenean viverra rhoncus pede. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Ut non enim eleifend felis pretium feugiat. Vivamus quis mi. Phasellus a est. Phasellus magna. In hac habitasse platea dictumst. Curabitur at lacus ac velit ornare lobortis. Curabitur a felis in nunc fringilla tristique. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem. Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu. In enim justo, rhoncus ut, imperdiet a, venenatis vitae, justo. Nullam dictum felis eu pede mollis pretium. Integer tincidunt. Cras dapibus. Vivamus elementum semper nisi. Aenean vulputate eleifend tellus. Aenean leo ligula, porttitor eu, consequat vitae, eleifend ac, enim. Aliquam lorem ante, dapibus in, viverra quis, feugiat a, tellus. Phasellus viverra nulla ut metus varius laoreet. Quisque rutrum. Aenean imperdiet. Etiam ultricies nisi vel augue. Curabitur ullamcorper ultricies nisi. Nam eget dui. Etiam rhoncus. Maecenas tempus, tellus eget condimentum rhoncus, sem quam semper libero, sit amet adipiscing sem neque sed ipsum. Nam quam nunc, blandit vel, luctus pulvinar, hendrerit id, lorem. Maecenas nec odio et ante tincidunt tempus. Donec vitae sapien ut libero venenatis faucibus. Nullam quis ante. Etiam sit amet orci eget eros faucibus tincidunt. Duis leo. Sed fringilla mauris sit amet nibh. Donec sodales sagittis magna. Sed consequat, leo eget bibendum sodales, augue velit cursus nunc, quis gravida magna mi a libero. Fusce vulputate eleifend sapien. Vestibulum purus quam, scelerisque ut, mollis sed, nonummy id, metus. Nullam accumsan lorem in dui. Cras ultricies mi eu turpis hendrerit fringilla. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; In ac dui quis mi consectetuer lacinia. Nam pretium turpis et arcu. Duis arcu tortor, suscipit eget, imperdiet nec, imperdiet iaculis, ipsum. Sed aliquam ultrices mauris. Integer ante arcu, accumsan a, consectetuer eget, posuere ut, mauris. Praesent adipiscing. Phasellus ullamcorper ipsum rutrum nunc. Nunc nonummy metus. Vestibulum volutpat pretium libero. Cras id dui. Aenean ut eros et nisl sagittis vestibulum. Nullam nulla eros, ultricies sit amet, nonummy id, imperdiet feugiat, pede. Sed lectus. Donec mollis hendrerit risus. Phasellus nec sem in justo pellentesque facilisis. Etiam imperdiet imperdiet orci. Nunc nec neque. Phasellus leo dolor, tempus non, auctor et, hendrerit quis, nisi. Curabitur ligula sapien, tincidunt non, euismod vitae, posuere imperdiet, leo. Maecenas malesuada. Praesent congue erat at massa. Sed cursus turpis vitae tortor. Donec posuere vulputate arcu. Phasellus accumsan cursus velit. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Sed aliquam, nisi quis porttitor congue, elit erat euismod orci, ac placerat dolor lectus quis orci. Phasellus consectetuer vestibulum elit. Aenean tellus metus, bibendum sed, posuere ac, mattis non, nunc. Vestibulum fringilla pede sit amet augue. In turpis. Pellentesque posuere. Praesent turpis. Aenean posuere, tortor sed cursus feugiat, nunc augue blandit nunc, eu sollicitudin urna dolor sagittis lacus. Donec elit libero, sodales nec, volutpat a, suscipit non, turpis.

3.3 Weitere Abschnitte ...

Nullam sagittis ...

4 Fazit und Ausblick

(...)

Bibliography

Clymo, R. S. (1970). The Growth of Sphagnum: Methods of Measurement. *Journal of Ecology*, 58(1):13–49. Publisher: [Wiley, British Ecological Society].